

Interview Questions — Java - RDBMS & Database Programming With JDBC

Total Questions: 41

1. What is JDBC and why do we need it in Java applications?

Hint: Think about how Java interacts with databases.

2. Explain the main components of JDBC architecture.

Hint: Focus on Driver Manager, Connection, and Statement.

3. In a typical Java application, how does JDBC connect to a MySQL database?
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4. What is the purpose of the MySQL connector JAR file when working with JDBC?

Hint: Consider how Java communicates with MySQL.

5. Describe the steps to establish a database connection using JDBC in Java.
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6. Write a short Java code snippet to connect to a MySQL database using `DriverManager.getConnection()`.

Hint: Remember to include the connection URL, username, and password.

7. How should you handle exceptions when establishing a JDBC connection?
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8. What is the difference between `Statement` and `PreparedStatement` in JDBC?

Hint: Think about performance and security aspects.

9. Why is `PreparedStatement` preferred over `Statement` for executing queries that include user input?

Hint: Consider SQL injection risks.

10. Write a Java code snippet using `PreparedStatement` to insert a new record into an 'expenses' table.
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11. How can you update or delete a record in a database using `PreparedStatement`?
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12. What is SQL Injection, and how does using `PreparedStatement` help prevent it?

Hint: Think about how user input is handled.

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13. How do you execute a SELECT query using JDBC and process the ResultSet to display results?
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14. Write a Java method that retrieves all expenses for a given user from the database and prints them in a formatted way.
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15. What is a ResultSet in JDBC? How do you traverse it?
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16. Explain how you would implement pagination and sorting when displaying results fetched from a database.
- Hint:** Consider using SQL LIMIT/OFFSET and ORDER BY.
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17. What is auto-commit mode in JDBC, and how does it affect transactions?
- Hint:** Think about when changes are saved to the database.
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18. How do you manually manage transactions in JDBC using commit() and rollback()?
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19. Write a code snippet to insert multiple expense records in a single batch using JDBC.
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20. Why is batch processing useful when inserting large numbers of records?
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21. If an error occurs during a batch insert, how can you ensure that none of the records are saved to the database?
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22. How does the try-with-resources statement help with resource management in JDBC?
- Hint:** Think about closing connections and statements automatically.
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23. What are the risks of not closing JDBC connections properly?
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24. Explain the difference between DatabaseMetaData and ResultSetMetaData in JDBC.
- Hint:** One gives info about the database, the other about the result set.
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25. How would you use exception handling to make your JDBC code more robust?
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26. How can AI tools like GitHub Copilot assist you in writing JDBC code templates, and what should you check before using its suggestions?
- Hint:** Think about code correctness and security.
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27. Describe the overall architecture of a mini project that integrates Java, JDBC, and SQL for an expense tracker application.
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28. How would you implement user registration and login functionality in a console-based Java application using JDBC?
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29. Write a Java method to add a new expense for a user, including input validation and error handling.
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30. How would you retrieve and display a monthly expense report grouped by category using JDBC and SQL?
- Hint:** Consider using SQL GROUP BY and date filtering.
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31. What is the MVC (Model-View-Controller) pattern, and how does it apply to your Smart Expense Tracker project?
- Hint:** Think about separating database logic, user interface, and business logic.
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32. If a user tries to delete an expense that doesn't exist, how should your application handle this scenario gracefully?
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33. Suppose your application is showing incorrect totals in the monthly report. What steps would you take to debug and fix the issue?
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34. How would you validate user input to prevent invalid data from being saved to the database in your expense tracker app?
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35. Describe how you would integrate an 'AI Tips' section that fetches budgeting advice from an external API like ChatGPT in your Java application.
- Hint:** Think about making HTTP requests and displaying responses.
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36. If you used Copilot or ChatGPT to generate JDBC code for your project, how would you verify that the generated code is correct and secure?
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37. Can you explain how exception handling and transaction management work together in JDBC to ensure data integrity?
- Hint:** Think about rolling back transactions when exceptions occur.
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38. In your mini project, how did you structure your code to separate database access logic from business logic and user interface?
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39. Write a Java function that updates multiple expense records in a transaction, ensuring that either all updates succeed or none are applied.

40. How would you use `ResultSetMetaData` to dynamically display column names and values when printing query results?

Hint: Think about iterating over columns in the `ResultSet`.

41. Suppose you need to support a different database (e.g., PostgreSQL) in your Java application. What changes would you need to make in your JDBC setup?
